1 Rationale and positioning with regard to the state-of-the-art

Driven by the many successes of (generative) AI in the last few years it has become more and more apparent that there is a need for reliable AI. This need manifests itself e.g. in the EU AI Act [2], and in the many conferences, workshops and special editions (e.g. [3], [1], [4]) on this topic. There are many ways in which the current research environment can contribute to this. In this project we will focus on the contributions related to *adversarial robustness* and interpretability through *disentanglement*.

For this we will use heterogeneous learning schemes. In particular, we will focus on *multi-view learning*. The reasons for this are twofold: firstly a lot of real-world applications are inherently presented as a multi-view problem; e.g. quality prediction in industry where each sensor provides a different set of features [31]. A straightforward, but naive approach is to concatenate the features of all views. However, this approach ignores the statistical properties of each view, and research (e.g. [22]) has shown that multi-view techniques that jointly optimizing different objectives can improve the performance drastically. A second reason to focus on multi-view learning is that it inherently causes a model to be more robust. Since each sample has multiple representations, if data from one view is corrupted (due to an attack, noise, ...) the information needed might still be present in the other views [37].

The arms race between adversarial attacks and defenses has been going on for several years now, where each time a defense system, that defends against most of the existing attacks is proposed, a new type of adversarial attack is able to fool the system. Intuitively it makes sense that using multiple types of defense methods improves the robustness of the entire system, since different attacks might be detected by different types of defense methods. It is therefore not surprising that ensembles of defense methods have been proposed (e.g. [44], [46]). However, as He et al. [18] notes, simply taking an ensemble of different defenses might not be sufficient to counter attacks. In a recent work, Wang et al. [45] introduced the use of multi-task learning to improve adversarial robustness, by introducing an extra learning tasks based on complementary robust labels. Furthermore, most research in adversarial learning is about unstructured data (in particular mostly on images), but structured data can be vulnerable as well [30, 19].

Interpretability of AI models can come in many forms [16]. An example are surrogate models such as SHAP [28] and LIME [35], which explain individual predictions of trained deep neural network models. In this work we will focus on interpretability of generative modelling using disentanglement [20], which measures how well individual components in the latent space match ground-truth factors (e.g. shapes, colors,...) that generated the data. This makes the model interpretable by itself and allows for carefully guiding the generation process to the desired outcome. State-of-the-art models for disentangled feature learning include InfoGAN [10], Restricted Boltzmann machines [34], β-VAE [20] and the recently proposed deep kernel PCA [43]. However few of these models are usable in a multi-view setting.

In the last years, deep learning neural network architectures have made for some major progress and impact in many exciting applications such as drug discovery [6], climate science [33] and text generation [8]. However, it also raises new questions on the fundamental capabilities and limitations. Recently, several kernel-based approaches have been developed that provide a deeper understanding and more solid theoretical foundation (e.g. [7], [40]), complementary to the powerful and flexible deep learning architectures. The research in this project will advance the connection between kernel-based and deep learning, and by doing so surpass the state-of-the-art in both fields.

In kernel methods, one usually optimizes the dual problem. The main reason being that this avoids having to explicitly defining the feature map by taking advantage of the kernel trick. However, when the dataset is large it can be more efficient to solve the primal problem. In order to still have a non-linear solution the feature map needs to be explicitly defined (e.g. [11]) or approximated (e.g. [41]). For the latter a sampling scheme is needed to determine landmark points. Some state-of-the-art schemes include DPPs [25] and RAS [15], but none of these extend readily towards multi-view learning. Moreover by investigating diverse sampling schemes, we also reduce the influence of potential bias in the model. This will increase fairness in the proposed multi-view methods. This is an important research avenue, since very little research has been done concerning fairness in a multi-view setting. With multi-view learning also comes a third scalability issue, namely the number of views. As an example, MV-LSSVM [22] training has a time complexity of $O(N^3V^3)$, hence as the number of views (V) is high (especially combined with a large number of datapoints (N)) the complexity will quickly increase. For multi-task learning this issue has been studied before resulting in approaches based on online learning (e.g. [13]), distributed models (e.g. [49]) and parallelization (e.g. [50]). Some studies include the concept of task relatedness (e.g. [17]) in order to decrease the number of tasks. However, it appears that scalability w.r.t. a large number of views in multi-view learning remains largely unexplored.

Currently there is not a lot of overlap in research between the domains of robustness, interpretability and multi-view learning (with a notable exception being the recent research interest in multi-modal explanations [36]). We expect our project to impact the current international research landscape and bring towards more attention and further research into these issues.

2 Scientific research objectives

The overall objective of this project to *enhance AI reliability through multi-view learning*. We will contribute to this overarching goal in several ways. Since disentangled representations make for more interpretable models, a first objective will be to achieve disentanglement in multi-view learning. A second improvement is by increasing adversarial robustness through multi-view and multi-task learning. Finally as scalability ensures that the systems can perform reliably in diverse and real environments, we will address several scalability issues. One way of dealing with scalability in terms of large datasets includes sampling methods which promote diversity, which in turn also reduces potential bias in the model.

Objective 1. Achieving disentangled representations in multi-view learning We aim at obtaining a deeper understanding of disentangled representations in multi-view learning both for shallow and multi-level architectures, and for deep learning architectures that relate in their dual forms to either symmetric or asymmetric kernel functions.

Objective 2. Using multi-view and multi-task schemes to improve robustness It was previously stated that multi-view models are inherently more robust to adversarial attacks than their single-view counterparts. The objective is to further study this property and to go a step further by using multi-view and multi-task techniques to improve adversarial robustness even more.

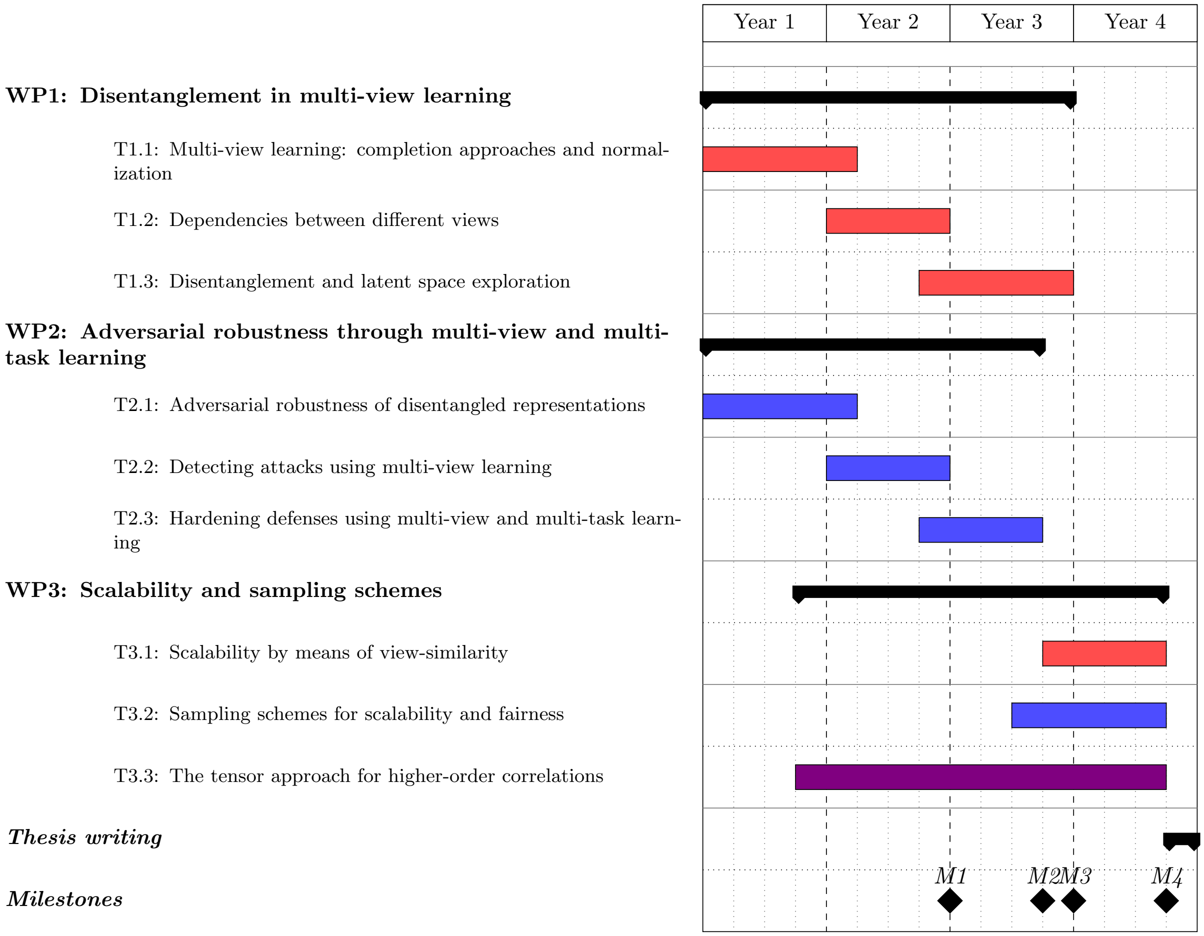

Figure 1: Work plan including division of labour between the two PhD candidates. The red and blue lines correspond to the PhD student primarily supervised by respectively PI Suykens and PI Houthuys. Task T3.3 will be worked on by both students.

Objective 3. Improving the scalability in a multi-view setting When using multi-view methods there is a scalability issue that needs to be taken into account. Not only do we want these methods to be scalable w.r.t. the number of inputs and the dimension of the input, but also the number of views. A high number of views can cause computational and memory issues, but it also introduces higher order correlations between them. The third objective is to develop methodologies to deal with these issues, using view-similarity, various sampling schemes and tensor based representations.

3 Research methodology and work plan

The three objectives above translate into three corresponding work packages where PI Suykens and PI Houthuys will take responsibility for WP1 and WP2 respectively. WP3 will be a shared responsibility. This responsibility includes project management tasks like overseeing the progress and handling administrative issues. PI Houthuys will be responsible for the overall project management. The two PhD students will work in parallel, where the student primarily supervised by PI Houthuys will focus mostly on issues of adversarial robustness and the student primarily supervised by PI Suykens on disentanglement.

All work packages relate to the general objective of increasing AI reliability through multi-view learning. This entails that WPs share common goals such as combining information from multiple views in efficient ways. All WPs furthermore will work on the same use cases, detailed in Section 3.4. To ensure optimal collaboration, the students will be (co)-supervised by both PIs and we will regularly (at least once every four months) organize a one-day progress meeting. At least once a year we will furthermore invite external researchers as a means to quickly get feedback on our progress.

Figure 1 represents the planned course of activities and division of labour over the 4-year period. It also shows the estimated timeline for the four main milestones:

- **M1:** A deeper understanding of the learning processes and robustness of multi-view models
- **M2:** Disentangled representation in multi-view learning for shallow and multi-level architectures
- **M3:** A robust hardening framework for multi-view learners
- **M4:** A thorough methodology to handle scalability issues presented by multi-view methods

3.1 WP1: Disentanglement in multi-view learning

This work package will work towards attaining the first objective, i.e. achieving disentangled representations in a multi-view setting. For this we will first need a deeper understanding of the learning processes behind multi-view models, which is the goal of tasks T1.1 and T1.2. Together with T2.1 and T2.2, these tasks will contribute to milestone M1. Task T1.3 dives in to the disentanglement property in RKM based multi-view models, both for shallow and multi-level architectures, obtaining milestone M2.

T1.1: Multi-view learning: completion approaches and normalization In RBM and deep Boltzmann machines one takes a probabilistic approach where the joint distribution in the visible and hidden units [21] directly relates to the energy function to be minimized, and normalized by the partition function. In this setting one can either train the model or generate new data from the trained model. In a non-probabilistic setting it is the question how one could either train or generate related to a same underlying function. In [32] this was done by employing a super-objective function. The same function is used then for training and generating, but different elements are known for training versus generating. This could therefore be seen as a completion problem, for which a part is given and the missing unknown part should be completed. Depending on what is given (either for training or for generating) different missing information is to be completed. We will study completion through super-objectives for deep kernel machines within a multi-view framework [5]. For RKM representations to KPCA one has the remarkable property that the sum of all terms in the objective function (i.e the regularized loss) is zero for the training problem. We will investigate how this is related to normalization aspects in a non-probabilistic setting and how this property can be further exploited for improved prediction and generative schemes. Furthermore, the completion and normalization methods will be extended to multi-view learning and multi-view multi-task learning.

T1.2: Dependencies between different views In [11] it has been shown of self-attention can be conceived as a modified version of kernel singular value decomposition within the least squares support vector machine framework. It involves an asymmetric kernel corresponding to two feature maps (related to queries and keys). It yields a regularized loss for training transformers to obtain low rank representations and efficient training in the primal. It will be investigated how such model formulations can be extended to multi-view problems. In the context of solving classification or regression problems the multi-view approach may also reveal dependencies between input variables or groups of input variables. In kernel singular value decomposition the dimensionality of the data in each of the data sources can be different. For having compatible feature maps a compatibility matrix can be introduced [39]. It will be studied for multi-view problems how one can optimally learn compatibility matrices. Several possible objectives can then be considered at this point, depending on whether one wants to analyse dependencies between different views or wants to obtain optimal classification performance.

T1.3: Disentanglement and latent space exploration In [43] new methods and models were recently proposed for Deep Kernel PCA, where the framework can either be used for training deep learning architectures (e.g. with convolutional feature maps) in primal form or with deep kernel representations in dual form. RKMs with one level [32] are characterized by an eigenvalue decomposition of the kernel matrix, such that each neuron in the neural network interpretation corresponds to one component, which are orthogonal to each other. For deep kernel PCA [43] it is also possible to obtain this property for multiple levels, for a deep hierarchical architecture. It leads to a deep eigenvalue decomposition analysis. This greatly enhances the interpretability of the model, and aims at obtaining a "deep white box", where the interpretation of each neuron is clear (e.g. when applying to an industrial process each neuron can correspond to a specific step in the process). The RKM models naturally possess this desirable property, while in other models such as VAE (variational autoencoders) one needs to additionally impose the disentanglement property in the latent space. In this proposal we will further investigate this towards multi-view learning. Kernel PCA and Deep Kernel PCA corresponds to a single data source. We will study deep multi-level architectures for multi-view learning with multiple data sources.

3.2 WP2: Adversarial robustness through multi-view and multi-task learning

Previous work (e.g. [45], [38]) have shown the potential of using heterogeneous learning schemes to increase adversarial robustness. This work package adds to the existing research in three major ways: (1) investigating the adversarial robustness of disentangled representations in a multi-view generation framework (T2.1), (2) proposing multiple ways to detect attacks using multi-view learning (T2.2) and (3) designing defenses based on multi-view and multi-task learning (T2.3). Together with T1.1 and T1.2, the first two tasks will contribute to milestone M1. Task T2.3 will result in milestone M3. In order to study defenses we will also need to study attacks, most of these will need to be designed specifically for the multi-view setting. In contrast to a lot of research into this topic, we will not limit ourselves to only image or other unstructured data but take into account structured data, like tabular data, as well.

T2.1: Adversarial robustness of disentangled representations Earlier work [32] proposed an RKM-based framework for multi-view generation (Gen-RKM) with disentangled feature learning. Since disentanglement has been shown to improve the robustness of VAEs against adversarial attacks ([47]), we will study whether this property holds for the multi-view RKM-based method as well. In order to do this we need to design appropriate adversarial attacks. For this we can start from the work of Kos et al. [24], which manipulates the latent representation. We believe that, using the same underlying methodology, we can design attacks that manipulate the latent variables of Gen-RKM. The main challenge will be to extend these attacks to the multi-view aspect of Gen-RKM. Once the attacks are developed we can start empirical investigations using commonly used datasets and robustness metrics tailored to generative models such as r-robustness [9].

T2.2: Detecting attacks using multi-view learning In this task we research the possibilities of using multi-view learning to detect adversarial attacks. The classification method proposed in [22] couples the view-specific error variables $\mathbf{e}^{[v]}$, which we expect can be used to detect poisoning attacks on a certain view. Another approach could be to look into view-specific predictions in order to detect evasion attacks. An advantage of this approach is that it is more widely applicable to all multi-view models that allow such predictions. We expect to be able to include these detection mechanisms into the multi-view methods without compromising the classification or clustering performance, and with only a slight increase in time complexity. We will test the method against several state-of-the art multi-view adversarial attacks. In a first stage we can use the recent work by Sun & Sun [37] which designed adversarial attacks specifically for multi-view image models. Since this imposes a strong restriction on the possible input, we will furthermore study how to extend attacks for a broader range of input types, such as tabular data (e.g. [19]) and text data (e.g. [14]), to the multi-view setting. For this we can take inspiration from [37] as well as from T2.1 where we did a similar extension to multi-view learning but for a generative task instead of a discriminative one.

T2.3: Hardening defenses using multi-view and multi-task learning In order to obtain an overall hardening defense for multi-view methods, we will look into multi-task learning schemes to combine defense mechanisms. Wang et al. [45] introduce this idea, but the performance of the proposed method heavily depends on the quality of the generated adversarial samples and robust labels. In order to improve on this we can use the idea of Tramèr et al. [44] and decouple the generation of adversarial samples from the model being trained, by using samples crafted on other pre-trained models. Furthermore, the task as described by Wang et al. is only one way of using the multi-task paradigm. We will investigate multiple ways to define the tasks, and not limit ourselves to adversarial training but look into other hardening and detection/denoising defences. Similarly to task T2.2 we will extend our methods to deal with more than image data alone.

3.3 WP3: Scalability and sampling schemes

In this work package we study scalability issues in the multi-view setting, i.e. related to the number of samples, dimension of the input and, in particular, the number of views. Tasks T3.1 and T3.2 tackle issues related to computational and memory complexity, while T3.3 is related to both the scalability and the increased likelihood of higher-order correlations when dealing with more than two views. Task T3.2 deals with sampling schemes which have an extra benefit of promoting fairness through diversity sampling. Together all tasks obtain milestone M4.

T3.1: Scalability by means of view-similarity In this tasks we study scalability of multi-view models related to a large number of views. For this we can take inspiration from techniques developed for multi-task learning and large class-set classification. E.g. Luckner [27] reduces the number of classifiers by using class similarity. Similarly we can study view-similarity. Note that since view-similarity and view-dependency are related, we might leverage insights obtained in T1.2. Since previous work has shown that multi-view classification is most useful when the views provide enough diverse information [22], we suspect that view-similarity can be used to decrease complexity when dealing with a large number of views.

T3.2: Sampling schemes for scalability and fairness A popular way to deal with issues of scalability in terms of large datasets in (single-view) kernel methods has been Nyström approximation [48]. The accuracy of such an approximation depends heavily on sampling schemes that select landmark points. Most of these sampling schemes are based on properties of the data distribution, which can be diverse among multiple views. We will look into recent sampling schemes such as Randomized Adaptive Sampling (RAS) [15], and extend these to the multi-view setting. The advantage of using schemes like RAS that promote diversity is that these also reduce potential bias and hence promote fairness of the model.

T3.3: The tensor approach for incorporating higher-order correlations When dealing with a large number of views, there is an increased likelihood that there exist higher-order correlations between them. Since most multi-view techniques rely on pairwise correlations, these higher-order relations are lost. Previous work (e.g.: [23], [42]) has shown great promise by using tensors to model these higher-order interactions. Moreover, Tao [42] also showed that the training complexity of these tensor models no longer depends on the number of views. Therefore, when proposing new models in WP1 and WP2 we will consider tensorized versions when appropriate, i.e. when the number of views is large and we suspect higher-order correlations to be important.

3.4 Benchmark data and use cases

Given the fundamental nature of the project, we will focus on publicly available data. This ensures that time spent on collecting and cleaning data is kept minimal and that we can compare our proposed solutions to other recent state-of-the-art methods.

Aside from the commenly used benchmark datasets (such as NUS-WIDE [12] and AwA [26]) we will use two specific use cases throughout the project. These use cases are specifically chosen to provide a variety of data types and scalability issues. Depending on the task we will consider generative (e.g. T2.1) or predictive (e.g. T3.2) problems. The exact learning problem (e.g. classification versus regression) will be decided based on the nature of the objective, the latest state-of-the-art methods we want to compare with and the available benchmark datasets. However note that in kernel-based models the classification and regression objectives usually only differ in small ways and hence the models can often be adjusted for the other task in a straightforward manner.

UC 1: Multi-sensor quality control in industry The AI4IM dataset[1] [31] consist of real-world and simulated data from an injection moulding manufacturing process. The data

[1] `https://github.com/lynnTM/AI4IM`

contains measurements from multiple sensors (temperature, pressure, etc.) and can be used to do quality prediction and fault detection. The dataset contains data from up to 7 sensors, and each measurement can be described through several feature sets. While the number of datapoints is limited, we can use this dataset to study scalability issues related to a large number of views.

UC 2: Large-scale video dataset The YouTube video games dataset [29] contains 120K video's represented by 13 views of visual, textual and auditory feature sets. Due to the high number of datapoints and views, this dataset is ideal to investigate the scalability issues discussed in WP3.

4 Analysis of Risks and Bottlenecks

The research goals of this project are both fundamental and ambitious, which lead to certain scientific risks. We mitigate these by ensuring alternative approaches and limiting the dependence of the research progress between the two students. The students can work on WP1 and WP2 independently from each other. WP3, and especially task T3.3, will benefit more from collaboration, but in worst case only one student will work on this task.

Since the students will work in different universities there is a risk of low collaboration and cohesion between the work packages. To mitigate these risks we have defined common use cases and the two PhD students will be jointly supervised by the two PIs and spent time in both labs. Furthermore, the students will regularly present their progress during progress meetings, and setting up a digital project work space (e.g. Slack) where they can communicate efficiently with each other and the two PIs. Furthermore, the two PIs have successfully collaborated in the past, and plan on collaborating further beyond this project as well.

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